Autonomous odor source localization using event-driven deep reinforcement learning and transformer-based feature evaluation

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Abstract

Dangerous chemical gas leaks in indoor ventilation environments pose serious threats to human health and safety. A mobile robot performing odor source localization (OSL) faces two major challenges: insufficient exploration leads to the robot being trapped in local optima and recirculation zones, and noise interference or communication delays cause partial sensor information loss. To solve these problems, this paper proposes a soft computing approach combining Transformer-based feature evaluation with an enhanced group relative policy optimization (EGRPO) in a prioritized colored Petri net (PCPN) framework. First, the OSL problem is modeled as a partially observable Markov decision process (POMDP), and the state space, action space, and dense rewards are designed to mitigate the sparse reward issue. The PCPN achieves state transitions through an asynchronous and concurrent operating mechanism. Subsequently, an attention-based Transformer encoder is utilized to extract time-series features and evaluate feature relevance in different search scenarios. The EGRPO algorithm is used to generate optimal decisions, improving the robot's OSL performance. Finally, in the same environments, computational fluid dynamics (CFD) simulation experiments demonstrate that the proposed method outperforms the baseline PPO, with a 14.4% increase in average reward and a 28.5% reduction in average search steps.

Keywords: Odor source localization, Deep reinforcement learning, Transformer encoder, Enhanced Group Relative Policy Optimization, Prioritized colored Petri nets.

1. Introduction

Hazardous gas leaks are a major environmental contamination source, posing significant risks to human health and security. Typical examples include chemical gas releases from factories[1],

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toxic gas dispersion[2], atmospheric pollution[3], and underground gas leaks[4]. It is crucial to rapidly and accurately locate the leakage source to minimize its environmental impacts. Existing OSL approaches can be broadly classified into two categories: static sensor networks and autonomous robotic searches using robots equipped with sensors[5]. Sensor network-based methods primarily fuse data readings from a group of sensing nodes, integrating them with prior information (e.g., meteorological data) to estimate unknown source parameters through Bayesian inference methods[6] or least-squares optimization of concentration differences[7]. However, sensor nodes are spatially fixed and subject to substantial measurement errors in complex environments, so extensive diffusion regions require a substantial number of sensors. Moreover, static sensor networks lack flexibility when environmental conditions change, as their spatial configuration needs to be redeployed and recalibrated.

A mobile robot equipped with chemical or other physical sensors can use data (e.g., gas concentrations) to locate an odor (or volatile chemical) source in an unknown environment. Compared to static sensor networks, the mobile robot provides active goal-oriented sensing, flexible coverage with reduced deployment cost, and strong adaptability to complex and dynamic environments, making it suitable for OSL tasks under realistic conditions. The autonomous search generally comprises three processes: finding, tracking, and localization[8][9][10]. The robot in low-concentration environments uses random exploration to find the odor plume. In the tracking phase, the robot utilizes the plume concentration distribution to locate the leakage source. In the localization phase, the robot declares the target location and terminates the search when it detects a plume concentration exceeding a predefined threshold.

Deep reinforcement learning (DRL) has recently emerged as a prominent research method for the mobile robot to perform OSL tasks[11][12]. Without requiring an accurate model of the environment, DRL can autonomously learn adaptive decision-making policies oriented toward long-term objectives directly from high-dimensional perceptual information[13]. Besides, DRL demonstrates considerable potential in sparse reward environments[14], balancing exploration and exploitation while preserving policy diversity to avoid premature convergence to local optima[15]. In OSL tasks, a robot in unknown environments without prior knowledge determines its current state through sensor measurements[16]. Therefore, the OSL task can be formulated as a sequential decision-making problem involving intermittent sampling, motion planning, and obstacle avoidance[17]. Under indoor ventilation conditions, unstable gas dispersion further complicates this process with high-dimensional states and nonlinear dynamics. DRL, with the capacity of learning features from complex sensory inputs and optimizing sequential decisions, is well suited for performing OSL tasks[18]. Lee et al.[19] designed Gaussian mixture model (GMM)

variance and GMM information gain to enhance the reward function and employed a deep Q-network for decision-making, successfully finishing the plume-tracking tasks. Nevertheless, this method only considered stable and obstacle-free environments. Inspired by this, Park et al.[20] combined particle filtering with GMM to compress high-dimensional particle distributions into low-dimensional features and applied deep deterministic policy gradient (DDPG) for localization of the plume leakage source, circumventing limitations of discrete action spaces. Chen et al.[21] used a deep Q-network (DQN) and convolutional neural networks (CNNs) to output action values, combining complete environmental maps and precise self-localization to solve the OSL problem. Li et al.[22] proposed a GRU-PPO algorithm for localizing decaying odor sources, achieving source localization in both obstacle-free and obstacle-present environments. However, the aforementioned studies did not sufficiently capture the strong correlation between wind field information and concentration distribution in dynamic environments. To address this limitation, Wang et al.[23] applied fuzzy inference to dynamically combine wind-based source probability maps with plume distribution maps, improving the balance between exploration and exploitation during the search process. Nevertheless, this kind of approach was restricted to stable wind fields. To further enhance the utilization of wind information in indoor turbulent environments, Gongora et al.[24] modeled the OSL problem using a Gaussian Markov random field (GMRF) that combined gas concentration and wind vectors into a joint probability distribution and introduced an energy function to estimate the source location. However, this approach is sensitive to noise and computational demands.

In current research, recurrent neural networks (RNNs) and long short-term memory (LSTM) networks have been applied to handle sequential problems in OSL tasks. However, RNN-based architectures tend to forget early inputs when dealing with long sequences, leading to the loss of temporal dependencies that encode historical information. The Transformer architecture, proposed by Google[25], provides a compelling alternative through its parallel input processing and superior capability in capturing long-range dependencies. Ni et al.[26] proposed a Scentformer model for the detection of odors based on the Transformer and CNNs. The CNNs were used to extract local spatial features and the Transformer was utilized to capture global temporal dependencies. However, this approach needed lots of parameter training and computational resources to improve experimental performance. Son et al.[27] integrated the global attention mechanism of the Transformer with convolutional networks, recursively predicting previous concentration distribution images from reverse-ordered temporal sequences of gas concentration maps to backtrack toward the source, effectively handling spatiotemporal sequence data. However, the model's architecture comprising three encoder layers and three decoder layers entailed 172,104 trainable

parameters. Such complexity made the model difficult to train. In addition, the full Transformer architecture usually required large amounts of training data as prior knowledge[28]. Devlin et al.[29] designed a Class token to extract features, reducing computational complexity. Unlike other inputs, Class token itself carried no useful information, so it did not exhibit preference toward any specific part of the sequence. This design avoided the manual selection preference inherent in conventional methods, such as focusing exclusively on the start or end of sequences and applying pooling operations. Inspired by the Class token, this paper designs a learnable State token placed at the beginning of the input sequence to achieve feature extraction. Meanwhile, this paper only uses two encoder layers as a feature extractor within an actor-only framework, with the total number of trainable parameters amounting to 54,296. The computational complexity of the proposed method is lower than that of the full Transformer architecture.

Traditional RL algorithms based on the actor-critic framework require a large number of trainable parameters, reducing the efficiency of decision-making. The Group Relative Policy Optimization (GRPO) algorithm[30], a variant of the proximal policy optimization (PPO) algorithm[31], completely removes the critic network used in traditional PPO. GRPO based on the actor-only framework does not require a large number of trainable parameters and computational resources. GRPO directly computes the advantage function using the mean and standard deviation of samples in the group, and employs group relative rewards to enhance training stability while reducing computational costs, enabling agents to adapt to various complex scenarios. Yu et al.[32] improved GRPO and proposed the decoupled clipping and dynamic sampling policy optimization (DAPO) algorithm, which incorporated four techniques: the clip-higher strategy, dynamic sampling, token-level policy gradient loss, and reward shaping. DAPO achieved an AIME50 score on the Qwen2.5-32B model using only 50% of the training steps, outperforming the GRPO algorithm. Although DAPO is an enhancement of GRPO specifically designed for text training in large language models and is not directly applicable to OSL tasks, its training techniques remain suitable for accelerating training and improving the stability of GRPO.

Under indoor ventilation conditions, fluctuations in wind speed and direction, along with interactions between wind and surrounding structures, generate vortices and recirculation zones[33]. These regions are closely associated with local concentration extrema. The current literature seldom pays attention to the strong relationship between local concentration extrema and recirculation zones under global wind field information, resulting in longer search times and lower success rates in OSL tasks. This paper proposes a Petri net to enhance the robot's exploration. Petri nets[34] provide a mathematical framework for describing and analyzing discrete event dynamic systems. As one of the earliest general-purpose models for concurrent processes, Petri nets have

been widely adopted in intelligent transportation systems[35], flexible manufacturing systems[36], and system-modeling methods[37] as an essential tool for studying asynchronous and distributed systems. Petri nets consist of four fundamental components: places, transitions, arcs, and tokens. The place represents system states or conditions, and the transition expresses events that trigger changes in those states. The arcs denote directed edges from places to transitions and from transitions to places. The token indicates the state's activation or resource count. This explicit separation between states and events provides the intrinsic decision-making mechanism for the system's autonomous operation. Compared with other discrete event modeling approaches, such as finite state machines[38] and behavior trees[39], Petri nets offer a clearer and more structured representation of how events drive system operation. In OSL tasks, Petri nets provide a graphical way to model search behaviors[40]. For example, places can denote operational modes such as downwind exploration or upwind exploitation, while transitions correspond to events such as switching between search strategies. The distribution of tokens across places reflects the robot's current state in real time, enabling transparent monitoring of the behavioral process. Colored Petri nets[41] extend classical Petri nets by incorporating color attributes to represent distinct properties, and are suitable to model the robot's concurrent states in OSL tasks. Since the initial position of the robot is random, the leakage source may not lie in the airflow. A priority-based search strategy is added to the Petri net to address the randomness of initial placements. In this study, a prioritized colored Petri net (PCPN) is proposed to enable the robot to make autonomous decisions according to different behavioral states, thereby enhancing both the efficiency and reliability of leakage source search.

Inspired by the above research, this study develops an autonomous source localization approach (TOS-EGRPO) with high efficiency and reliability by combining PCPN with State token for feature extraction and an enhanced GRPO algorithm. The main contributions of this paper are as follows:

(1) An event-driven control method using PCPN is designed to strengthen the robot's autonomous search ability. This approach uses the graphical structure of places and transitions to reason about concurrent robot states, avoiding unnecessary decision events and achieving adaptive state transitions through the asynchronous and concurrent operating mechanism.

(2) A Transformer encoder architecture integrated a State token is designed to process noisy observation sequences and compensate for state information loss by extracting time-series features, enhancing the algorithm's robustness in partially observable environments.

(3) A task-specific enhanced GRPO algorithm is proposed. Modeling the OSL task as a POMDP, we develop the reward learning strategy including decoupled clipping, dynamic sam-

pling, and reward reshaping mechanisms to enhance decision-making.

(4) In the CFD-simulated 2D indoor plume environments, the proposed method demonstrates higher success rates and fewer search steps compared to DQN and PPO, particularly under the conditions involving long observation sequences and partial state information loss.

The remainder of this paper is organized as follows: Section 2 presents behavioral decision-making based on PCPN in OSL. Section 3 models plume tracking behavior as POMDP. Section 4 describes the TOS-EGRPO algorithm. Section 5 presents the experimental results and discussion. Section 6 draws the conclusions.

2. Problem formulation

2.1. Plume dispersion model

In an indoor space with airflow, gas diffusion often occurs in an irregular manner. Therefore, a SST $k-\omega$ model (Shear Stress Transport $k-\omega$) is employed in the simulation. The model simulates turbulence by solving the governing equations for turbulent kinetic energy (ki) and specific dissipation rate (w), utilizing the turbulent viscosity coefficient $u_t = (\rho * ki)/w$ for closure. This paper proposes a time-varying periodic source to model the fluctuating leakage process over time. The formula is as follows:

$$v(t) = v_0[1 + k\sin(2\pi f t)] \tag{1}$$

where $v(t)$ represents the time-varying release rate of the plume source in m^3/s, $v_0 = 0.4m^3/s$ is the initial release rate, $k = 0.5$ is the dimensional fluctuation ratio, $f = 1Hz$ is the fluctuation frequency, and t is the physical time in seconds. The argument $2\pi f t$ is dimensionless, ensuring dimensional consistency.

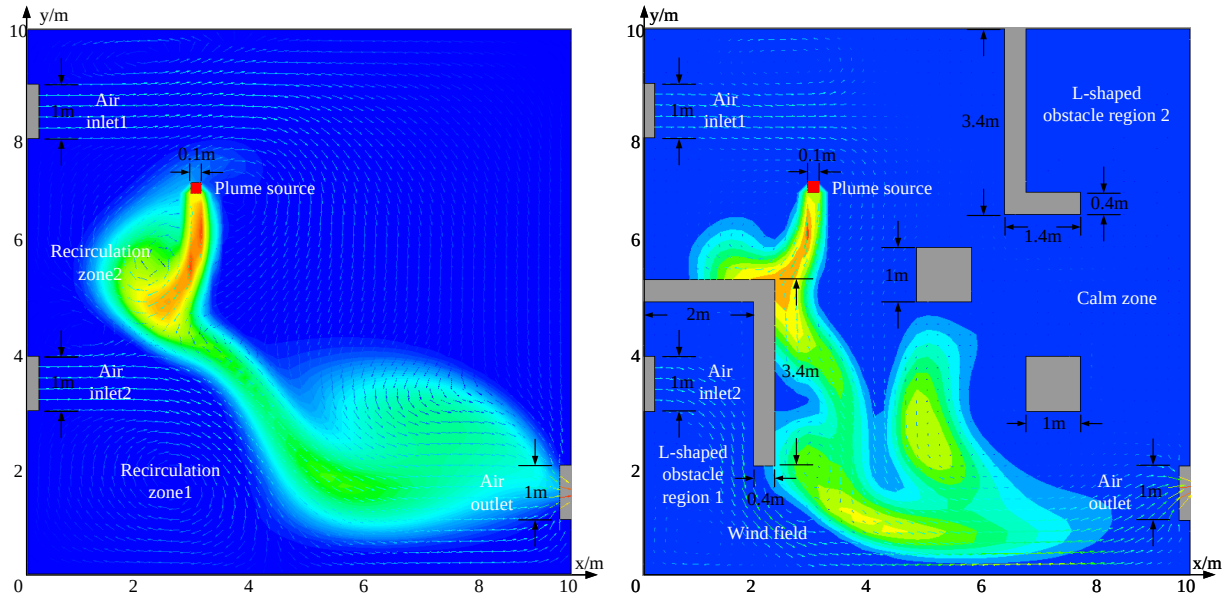
This study employs the Fluent module in ANSYS to simulate gas diffusion in a 2D indoor ventilation environment. The boundary conditions and solution models for CFD simulation are shown in Table 1 and Table 2. Gas concentration levels are visualized using a color gradient ranging from light green to red, while the wind field is represented by arrowed vectors. The indoor obstacle-free environment is defined as a 10×10 m square domain, featuring two inlets (each 1 m wide) on the left boundary and a single 1 m-wide outlet on the right boundary. The gas release source, modeled as a 0.1×0.1 m square, is positioned at coordinates (3,7) in the domain and emits gas along the negative y-axis direction, as shown in Fig. 1(a). For the environment with obstacles, two square obstacles (each with a 1 m side length) and two L-shaped obstacles are introduced to create more complex airflow conditions. Their placement divides the environment into wind field zones and calm zones, as shown in Fig. 1(b).

Table 1. CFD simulation: boundary condition setup.

Boundary type	Configuration
Source	Mass flow inlet
Obstacle	No-slip wall
Wall	Zero diffusive flux
Air inlet	Velocity inlet
Air outlet	Relative pressure outlet

Table 2. CFD simulation: solution model setup.

Category	Model/Method
Viscous Model	SST k-omega[42]
Species Model	Species transport
Mixture material	Air and alcohol
Velocity-Pressure Coupling	SIMPLE method
Spatial discretization	Second-order upwind
Time type	Transient
Wall Function	Standard wall function



(a) Obstacle-free environment

(b) Environment with obstacles

Figure 1. Experimental simulation environments.

2.2. Behavioral decision-making based on PCPN in OSL

Event-based control refers to a paradigm where learning updates are triggered by specific events rather than fixed time intervals[43]. In the traditional RL methods, updates typically occur at fixed time intervals and after completing a specific number of steps. In contrast, event-driven control adapts its learning schedule according to crucial events. This approach can be advantageous in unpredictable and dynamic environments by avoiding numerous ineffective and unnecessary decision events, thereby improving training efficiency and algorithmic performance. This study uses a colored Petri net to implement this paradigm. During downwind exploration and upwind exploitation, the TOS-EGRPO algorithm is invoked to achieve robot-environment interactions. This design alleviates reward sparsity caused by ineffective interactions and ensures that rewards obtained from each interaction are directly linked to the robot's valid choices in key decisions.

Hazardous gas tends to accumulate at outlets where robot deployment poses higher risks, so this study initializes the robot at air inlets or other areas. A priority-based search strategy is adopted to address the randomness of initial placements. The plume searching behaviors are prioritized in descending order as follows: source declaration, recirculation zones detection and escape, upwind exploitation, and downwind exploration. The priority rules are configured as: 1 for Pmax, 2 for Plow1, 3 for Plow2, and 4 for Plow3. The PCPN is defined as follows:

$$PCPN = (P, T, F, \Sigma, C, G, \Pi) \tag{2}$$

with,

- $P = \{p_1, p_2, ..., p_n\}$: set of places,
- $T = \{t_1, t_2, ..., t_n\}$: set of transitions,
- $F \subseteq (P \times T) \bigcup (T \times P)$: set of directed arcs for connecting places and transitions,
- $\Sigma = \{c_1, c_2, ..., c_n\}$: set of colors,
- $C : T \bigcup P \rightarrow \Phi(\Sigma), \Phi(\Sigma) \subset \Sigma$ is the color function,
- $G : T \rightarrow \{0, 1\}$ is the transitions guard function,
- $\Pi : T \rightarrow IN$ is the priority function, which assigns a priority level to each transition, where low value corresponds to high priority.

The plume search model based on PCPN is shown in Fig. 2 The initial state p_1 contains one red token, one green token, and two blue tokens. The blue tokens are triggered in key events such as downwind exploration and upwind exploitation.

Table 3. Meanings of places and transitions in PCPN model.

Place(p_n)	Meanings	Transition(t_n)	Meanings
p_1	No plume	t_1	$C < C_t$
p_2	Downwind exploration	t_2	$\left(\sigma_\phi > \phi_{thr}\right) \wedge \left(\bar{U} < U_{\min}\right)$
p_3	Escape behavior	t_3	Complete escape
p_4	Update concentration	t_4	$C_t < C < C_{\max}$
p_5	Upwind exploitation	t_5	$C \geq C_{\max}$
p_6	Plume source declaration	t_6	$d < d_{thr}$
p_7	Obstacle avoidance	t_7	$d \geq d_{thr}$
p_8	Update position	t_8	Complete obstacle avoidance
p_9	Evaluate the group relative advantage	t_9	Sample for state information
p_{10}	Choose the optimal policy	t_{10}	$\arg\max R$

The semantics of places and transitions are summarized in Table 3, where C_t denotes the concentration threshold for plume finding, and d represents the relative distance between the robot and boundaries or obstacles, calculated by Euclidean distance. $C_{\max}$ and d_{thr} represent the concentration threshold for confirming the plume source and the safety distance threshold, respectively. Δv and $\Delta\phi$ denote the variations in wind speed and wind direction angle, respectively. Each time the robot executes an action, it samples the wind vector $V_i = (v_i, \phi_i)$ and retains the most recent N samples. The average wind speed $\bar{U}$ and the standard deviation of wind direction angles σ_ϕ are then used to determine whether the robot is in recirculation zones, calculated as follows:

$$\bar{U} = \frac{1}{N}\sum_{i=1}^{N} |v_i| \tag{3}$$

$$\sigma_\phi = \sqrt{\frac{1}{N}\sum_{i=1}^{N} (\phi_i - \bar{\phi})^2} \tag{4}$$

where ϕ_{thr} represents the wind direction angle variation threshold in recirculation zones, and $U_{\min}$ denotes the minimum wind speed.

During the initial plume search phase, if the robot does not detect the plume, transition t_1 is triggered to initiate downwind exploration. At the same time, red and blue tokens move from p_1 to p_2. The robot then uses $\bar{U}$ and σ_ϕ to determine whether it is trapped in recirculation zones. If so, transition t_2 is activated to execute escape behavior, and red token moves from p_2 to p_3. Because

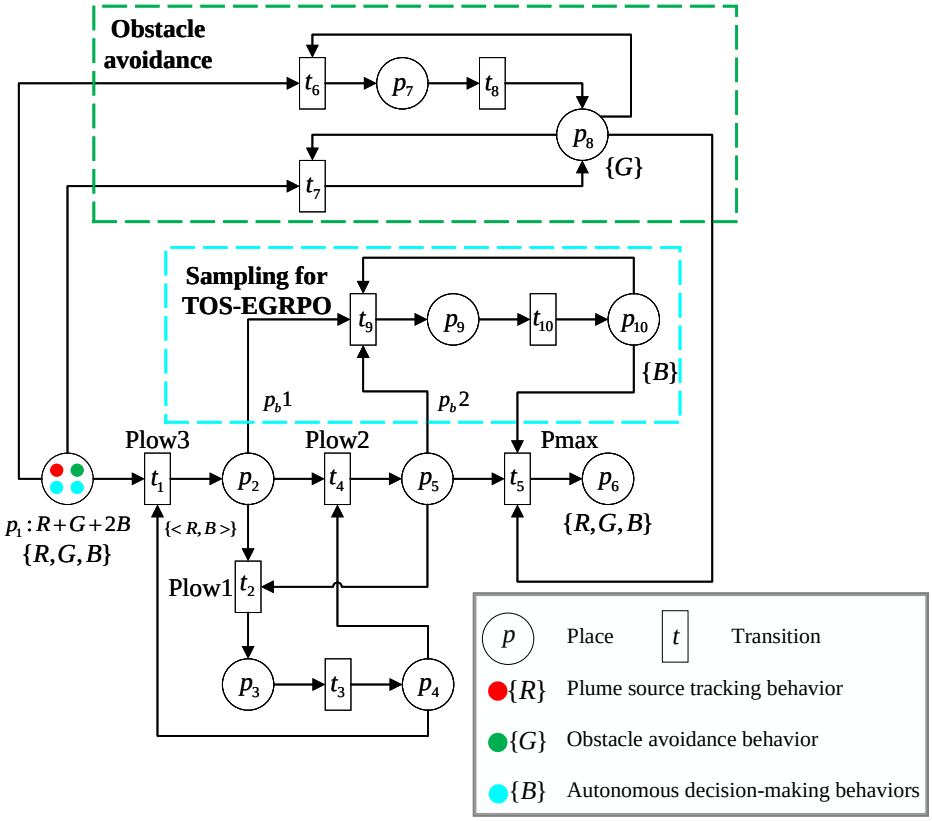


Figure 2. Odor source localization mechanism based on PCPN. With all arc weights being 1, three colored tokens denote concurrent behaviors that operate without interference. Search behaviors are executed in the order of priority: source declaration, recirculation zones detection and escape, upwind exploitation, and downwind exploration.

recirculation zones are characterized by low wind speed, the robot trapped in one can move along the wind speed gradient toward the direction of maximum wind speed until the wind direction variation stabilizes. This event triggers transition t_3 and moves to p_4. Upon plume detection by the robot, transition t_4 is activated to initiate upwind exploitation toward the maximum plume concentration, with red and blue tokens moving from p_2 to p_5. When the plume concentration exceeds threshold $C_{\max}$, transition t_5 is triggered to confirm the source location, and the red token shifts from p_5 to p_6. Throughout this process, obstacle avoidance behavior and autonomous decision-making based on the TOS-EGRPO algorithm proceed concurrently—represented by the green and blue tokens moving from p_8 and p_{10} to p_6, respectively.

2.3. Plume tracking process based on POMDP

Although the OSL problem can be formulated as a MDP with unknown transition probabilities [20], the various interference factors in real-world environments often result in partially observable states. Therefore, the problem is modeled as a discrete POMDP [40], which better reflects practical plume tracking scenarios.

To improve the feasibility of the CFD simulation calculations, this paper models the search area as a square grid network. Based on the defined grid size, the environment is divided into M grid cells, as shown in Fig. 3. The vector $C = [c_1, c_2, c_3, ..., c_M]$ defines the robot's position in the network, where $M = n \times n$, and n denotes the cell count per row and column. Moreover, $c_i = (x_i, y_i)$ represents the search location of the robot in the environment, where $i \in [1, M]$. Since the position of the gas leakage source is unknown to the robot, any one of M grid cells may potentially represent the plume source position.

2.3.1. Observation space

The observation space of the robot comprises both self-state information and environmental perception data, the latter acquired through onboard sensors. Sensors are assumed to update observations at fixed time intervals. Referring to Formula (1), the gas release rate reaches its peak value after T seconds during each emission period. The normalized observation space at time t is defined as follows:

$$\bar{s} = \left[\bar{x}, \bar{y}, \bar{q}, \bar{v}, \overline{\phi}, \bar{d}\right] \tag{5}$$

$$\bar{s}_i^t = \begin{bmatrix} C_{i-(n-1)}(\bar{s}), C_{i+1}(\bar{s}), C_{i+(n+1)}(\bar{s}), \\ C_{i-n}(\bar{s}), C_i(\bar{s}), C_{i+n}(\bar{s}), \\ C_{i-(n+1)}(\bar{s}), C_{i-1}(\bar{s}), C_{i+(n-1)}(\bar{s}) \end{bmatrix} \tag{6}$$

where C_i represents the ith grid's position in the search area. The vector $\bar{s}$ denotes the normalized observations, including the robot's position $(\bar{x}, \bar{y})$, gas concentration $\bar{q}$, wind speed $\bar{v}$, wind direc-

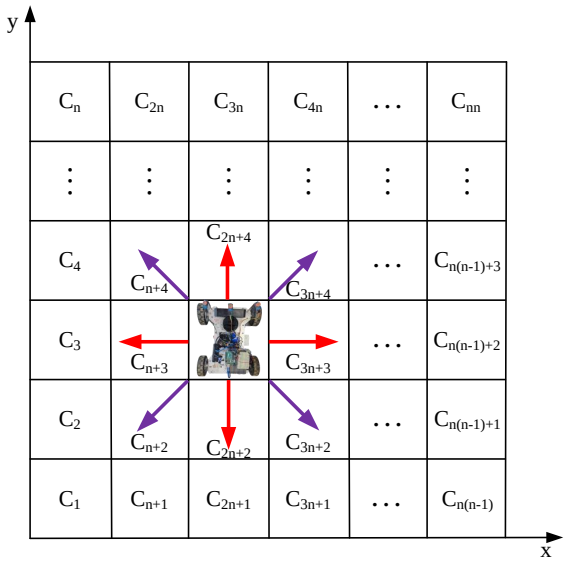


Figure 3. Search area and eight motion directions of robot.

tion $\bar{\phi}$ and the relative distance $\bar{d}$ to boundaries or obstacles. To speed up training of the neural network, each observation dimension is normalized to the range $[-1, 1]$. The processing procedure for position information is as follows:

$$\begin{cases} \bar{x} = 2\frac{x}{L} - 1, \ x \in [0, L] \\ \bar{y} = 2\frac{y}{L} - 1, \ \mathrm{y} \in [0, L] \end{cases} \tag{7}$$

where x and y represent the robot's horizontal and vertical coordinates in the grid, respectively; L is the side length of the square search area. The processing of environmental perception information proceeds as follows:

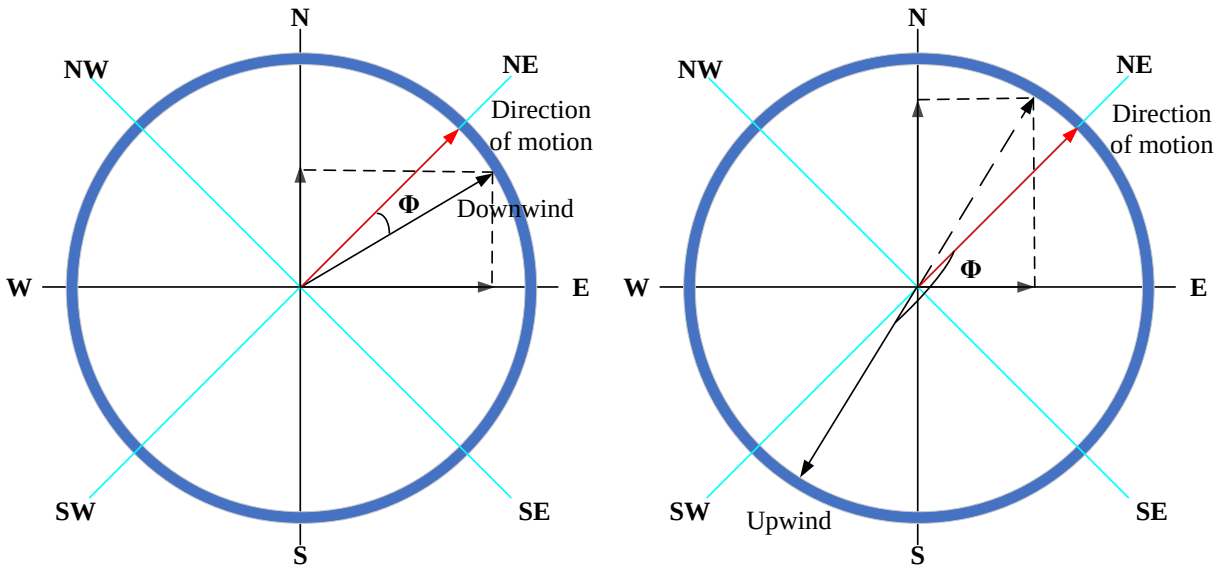
$$\begin{cases} \bar{q} = 2q - 1, \ q \in [0, q_{\max}] \\ \bar{v} = 2\frac{v}{v_{\max}} - 1, \ v \in [0, v_{\max}] \\ \bar{\phi} = \frac{\phi}{\phi_{\max}} - 1, \ \phi \in [0{,}360] \\ \bar{d} = 2\frac{d}{\sqrt{2}L} - 1, \ d \in [0, \sqrt{2}L] \end{cases} \tag{8}$$

where $q_{\max}$ represents the maximum plume concentration detected by the robot from its initial position to the source; $v_{\max}$ represents the maximum wind speed measured by the anemometer throughout the entire search process; $q_{\max}$ and $v_{\max}$ serve as empirical values that need to be adjusted in response to environmental changes.

The normalized observation space is represented as:

$$\bar{S} = [\bar{s}_1^t, \bar{s}_2^t, \bar{s}_3^t, \cdots, \bar{s}_M^t] \tag{9}$$

The robot can move in eight directions on the grid. The action space A comprises these possible movement directions, as shown in Fig. 3. When the robot moves along the horizontal and vertical axes, it has a fixed step size of N, with a fixed step of $\sqrt{2}N$ when going along the diagonal. The relationship between A and wind direction is shown in Fig. 4.



(a) Downwind exploration

(b) Upwind exploitation

Figure 4. Relationship between movement direction and wind direction.

2.3.2. Reward function

The design of reward functions involves the following four aspects: (1) It can solve the sparse reward problem in the OSL task; (2) It can guide the robot to rapidly approach the plume; (3) It can help the robot to avoid boundary violations and obstacle collisions; (4) It can minimize the number of search steps required. Therefore, the reward function primarily consists of six sub-reward functions: concentration exploration reward R_1, concentration exploitation reward R_2, declaration reward R_3, obstacle avoidance reward R_4, time step penalty R_5, and wind-guided reward R_6.

R_1 is designed to encourage the robot to rapidly locate the plume. R_2 aims to promote fast and accurate source localization while avoiding convergence to local extrema. R_3 is assigned to penalize behaviors of boundary violations and obstacle collisions, and to reward successful source declaration. The three reward functions are formulated as follows:

$$R_1 = \begin{cases} 0.8, & C \geq C_t \\ 0, & C < C_t \end{cases}, \quad R_2 = \begin{cases} 2\left(p_{\max} - p_t\right), & C \geq C_t \\ p_{\max} - p_t, & C < C_t \end{cases} \tag{10}$$

$$R_3 = \begin{cases} -200, \text{ beyond the boundary} \\ -10, \ d < d_{thr} \\ 200, \ C \geq C_{\max} \end{cases} \tag{11}$$

where $p_{\max}$ is the maximum concentration among the eight directions surrounding the robot; p_t represents the concentration at the robot's own location at time t.

R_4 prevents the robot from running outside the experimental boundaries and colliding with obstacles. R_5 encourages the robot to track the plume source with fewer search steps. A hierarchical penalty mechanism is introduced to better guide the robot in actively avoiding boundaries and obstacles, the specific mathematical form is:

$$R_4 = \begin{cases} 0, \ if \ d \geq d_{thr} \\ -\alpha \left(\frac{1}{1+e^{10\left(0.8-\frac{d}{d_{thr}}\right)}} \right), \ if \ d < d_{thr} \end{cases}, R_5 = -0.1 \tag{12}$$

where α defines the maximum penalty factor to control the severity of penalties.

R_6 is determined by concentration and wind direction: it encourages downwind exploration in low-concentration regions and promotes upwind exploitation in high-concentration regions. It is defined as follows:

$$w = \tanh\left[10\left(C - C_t\right)\right] \tag{13}$$

$$R_6 = 1.2 \times w \times \frac{v}{v_{\max}} \times \cos\left(\Delta\phi\right) \tag{14}$$

where $\Delta\phi$ represents the angle difference between the robot's moving direction and the wind direction, and the allowable range of it is $[0, 180]$. $v/v_{\max}$ indicates the normalized wind speed to improve comparability of wind intensity across different scenarios.

In summary, the rewards in the search process are composed of the above six reward functions. The overall reward named R_{total} under k steps search is mathematically expressed as:

$$R_{total} = \sum\nolimits_k \left(R_1^k + R_2^k + R_3^k + R_4^k + R_5^k + R_6^k \right) \tag{15}$$

2.3.3. State observation information loss and noise interference

The robot's observation space is related to information loss because of delays and interruptions in sensor communication. We incorporate an observation state information loss rate p into the experimental environment. Besides, this study designs a variable *random* with a value in the range of 0 to 1 to represent random probability. When *random* is less than p, the observation state information at the current time step is considered lost and replaced by the information from the

previous time step. Specifically, as shown in the following formula:

$$\bar{s}_t = \begin{cases} \bar{s}_t, \ random > p \\ \bar{s}_{t-1}, \ random \leq p \end{cases} \tag{16}$$

Existing studies commonly incorporate normally distributed noise to represent sensor errors[44], while rarely accounting for the relationship between noise intensity and signal strength. Given that the diffusion of gas molecules in a turbulent flow follows a Poisson point process[45], this study models sensor noise using Poisson-distributed random variables, which more accurately reflect real-world sensing conditions. Accordingly, Poisson noise is added to the normalized observation space to simulate realistic measurement interference, as expressed by the following equation:

$$\bar{s}_t^{noise} = clip(\bar{s}_t + noise, 0, 1) \tag{17}$$

where $clip(x, a, b)$ is to truncate x in $[a, b]$; $noise$ indicates noise sampled from $Noise$. The specific mathematical form is

$$Noise = C\frac{Poisson(\lambda) - \lambda}{\lambda} \tag{18}$$

$$\sigma \approx \frac{C}{\sqrt{\lambda}} \tag{19}$$

where λ represents the expected value of the Poisson distribution with a fixed value of 100; C denotes the noise amplitude scaling factor to limit the maximum noise amplitude with a fixed value of 0.15; σ represents the standard deviation with a calculated value of 0.015. By the normal approximation, $|noise| \leq 3\sigma \approx 0.045$ indicates that the probability of Poisson noise fluctuations is 99.73%. The noise standard deviation is 0.015. After 20 dB attenuation of the signal, i.e., when the signal amplitude is reduced to 0.1, the noise accounts for approximately 15% of the signal amplitude.

3. Learning strategy based on TOS-EGRPO

3.1. Enhanced GRPO algorithm

In the context of a POMDP, the GRPO algorithm is a modified local policy search method derived from PPO. GRPO incorporates meaningful learning signals into its estimated group relative advantage, which simplifies the computational process and accelerates training. The objective function of GRPO is as follows:

$$\hat{A}_{i,t} = \frac{R_i - \text{mean}(\{R_i\}_{i=1}^G)}{\text{std}(\{R_i\}_{i=1}^G)} \tag{20}$$

$$J_{GRPO}(\theta) = E\left[\sum_{t=0}^{T} \min\left(r_t(\theta)\hat{A}_{i,t}, \text{clip}\left(r_t(\theta), 1-\varepsilon, 1+\varepsilon\right)\hat{A}_{i,t}\right)\right] - \beta \mathrm{D}_{KL}\left[\pi_\theta \| \pi_{old}\right] \tag{21}$$

$$r_t(\theta) = \frac{\pi_\theta\left(a_t|s_t\right)}{\pi_{\theta_{old}}\left(a_t|s_t\right)} \tag{22}$$

where $r_t(\theta)$is the importance sampling factor that measures the relative change between new and old policies, and the clip function constrains the update in a trust region, enabling more cautious policy updates. $\hat{A}_{i,t}$ is the advantage calculated based on relative rewards of the outputs inside each group only. For each state-action pair (s, a), the behavior policy π samples a set of G action-response rewards $\{R_i\}_{i=1}^{G}$. D_{KL} denotes the KL divergence, and β is the weight of the KL penalty term, which is employed to balance the magnitude of policy updates against policy stability.

To improve computational efficiency, the KL divergence term is removed from GRPO and we fundamentally reformulate the optimization problem from token generation to a robot control-oriented POMDP.. Three training techniques from DAPO are applied simultaneously in the reinforcement learning framework: the clip-higher strategy, dynamic sampling, and overlong reward shaping. Different from the DAPO algorithm, we introduce task-specific constraints. These novel designs directly address key challenges in OSL, such as reward sparsity and ineffective physical interactions. The objective function for optimization is redefined as follows:

$$\begin{aligned} &J_{EGRPO}\left(\theta\right) = E_t\left[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o_i|}\sum_{t=1}^{|o_i|}\min\left(r_{i,t}(\theta)\hat{A}_{i,t}, clip(r_{i,t}(\theta), 1-\varepsilon_{low}, 1+\varepsilon_{high})\hat{A}_{i,t}\right)\right] \\ &s.t\ 0 < |\{o_i | is_equivalent(a, o_i)\}| < G,\ \{\tau \in \text{Episodes} | d_T(\tau) = 1\} \end{aligned} \tag{23}$$

where o_i represents that for each observation state s in an episode, an action group $\{o_1, o_2, ..., o_G\}$ is sampled by the old policy π_{old}, and $|o_i|$ is the observation length. $\{o_i | is_equivalent(a, o_i)\}$ defines a set composed of all objects o_i to satisfy the condition, and where a is the sampled action. $\hat{A}_{i,t}$ represents the group relative advantage, defined consistently with GRPO. $r_{i,t}(\theta)$ is the modified importance sampling ratio, whose mathematical formulation is given by:

$$r_{i,t}(\theta) = \frac{\pi_\theta(o_{i,t}|s, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t}|s, o_{i,<t})} \tag{24}$$

$\left(1-\varepsilon_{low}, 1+\varepsilon_{high}\right)$ separates the upper and lower limits of the clip range. The clip mechanism, which suppresses the probabilities of low-likelihood actions, allows the model to maintain existing high-probability actions and prevents a sharp decline in policy entropy. $0 <$

$|\{o_i|is_equivalent(a, o_i)\}| < G$ ensures that not all of the outputs in a group are either 0 or 1, thereby preventing the group from having zero advantage. The purpose is to avoid the issue of zero advantage leading to gradient-free policy updates to achieve dynamic sampling. In OSL tasks, dynamic sampling enables the robot to focus on both positive signals (e.g., successful localization) and negative signals (e.g., boundary crossing and obstacle collision) during training. At the same time, it encourages the robot to discard trajectories with negligible advantages (e.g., being trapped in the local concentration extremum). $\{\tau \in \text{Episodes} \mid d_T(\tau) = 1\}$ represents the overlong reward shaping, and its role is to eliminate the trajectories that have been truncated due to reaching the batch limit. $d_T(\tau) = 1$ indicates the robot has naturally ended the round in the environment. In OSL tasks, overlong reward shaping encourages the robot to more quickly learn to move toward the concentration gradient. This method reduces the loss of training signal caused by excessive or inefficient exploration, thereby enabling the robot to locate the gas leakage source in fewer search steps.

3.2. TOS-EGRPO algorithm

This study uses the State token to extract different kinds of time-series features from inputs. Since OSL tasks involve strict temporal dependencies, this paper incorporates positional encoding into the input sequence. To encode relative positions in the sequence, sinusoidal functions at different frequencies are employed. For an input sequence $X \in R^{n\times d}$, a positional embedding matrix $P \in R^{n\times d}$ (of the same shape) is created so that the output is $X + P$. The entries in row i, columns $2i$ and $2i + 1$ are as follows:

$$PE_{pos,2i} = \sin\left(\frac{pos}{10000^{2i/d}}\right) \tag{25}$$

$$PE_{pos,2i+1} = \cos\left(\frac{pos}{10000^{2i/d}}\right) \tag{26}$$

The State token serves as a learnable embedding that is concatenated with the other input state embeddings. It interacts with other embeddings through the multi-head self-attention (MSA) mechanism to extract representative features of the entire state sequence. Since the normalized state inputs share a uniform feature length and similar properties, it is difficult for the Transformer encoder to distinguish between them. Therefore, this study adds a segment embedding to each input state embedding to differentiate these states. The State token consists of a state embedding X_s and a learnable segment embedding E_s. The Transformer encoder module, including the State token and positional encoding, is processed through a feedforward neural network (FNN) followed by layer normalization (LN), as illustrated in Fig. 5.

The mathematical formulation of the feature extraction process is as follows:

$$z_0 = [X_S; S_A; S_B; S_C; S_D] + [E_S; E_A; E_B; E_C; E_D] + [PE_0; PE_1; PE_2; PE_3; PE_4] \tag{27}$$

$$z'_n = LN\left(z_{n-1} + MSA\left(z_{n-1}\right)\right), \quad n = 1, \ldots, N \tag{28}$$

$$z_n = LN\left(z'_n + FNN\left(z'_n\right)\right), \quad n = 1, \ldots, N \tag{29}$$

After being processed by N encoding layers, the model yields the final output:

$$z_n = [z_s, z_1, z_2, ..., z_n], \quad n = 1, \ldots, N \tag{30}$$

The output at the first position (corresponding to State token denoted as vector z_s) is extracted as a feature representation that aggregates global information from the entire state observation sequence. This feature vector is then fed into the actor network to generate a probability distribution over actions.

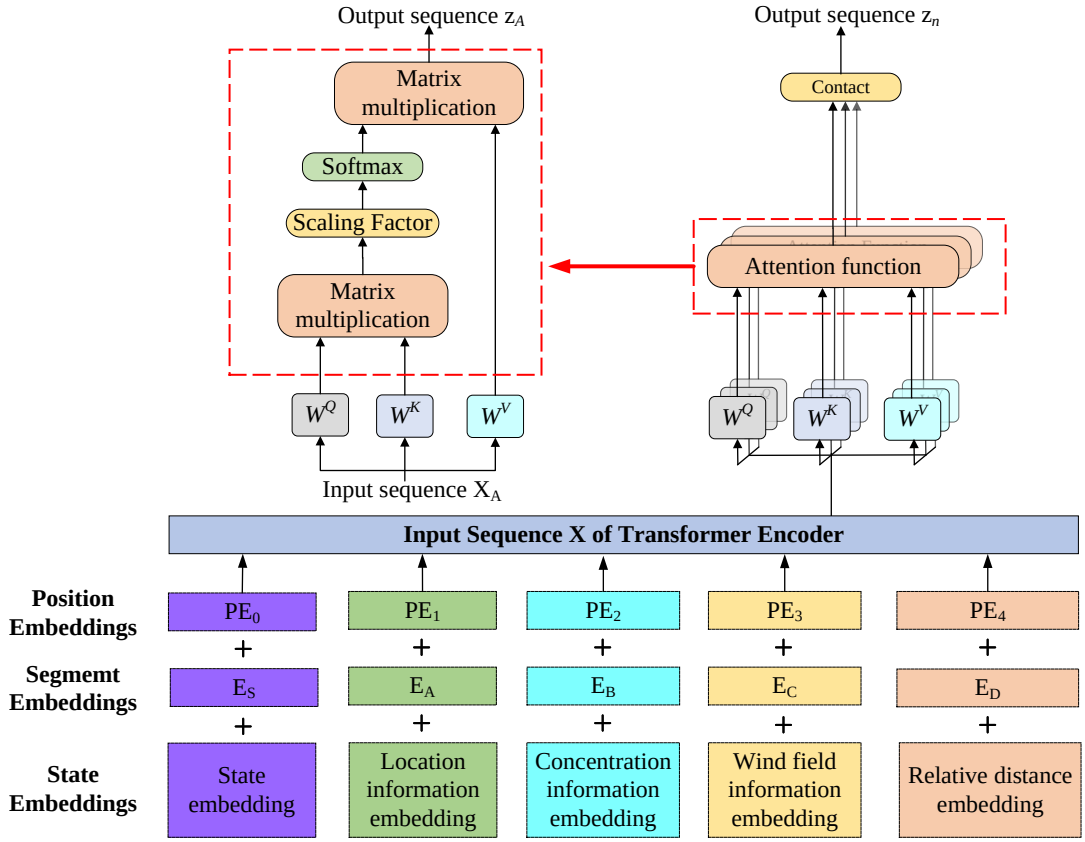


Figure 5. Attention mechanism in the Transformer encoder.

The TOS-EGRPO network architecture consists of three primary components: a Transformer encoder module, an actor network, and an output layer, as illustrated in Fig. 6. Firstly, the Transformer encoder with MSA automatically extracts hidden features from different types of information, while the State token captures sequential characteristics from historical observation data and reorganizes the resulting feature vectors. Subsequently, the extracted feature vectors are fed into the enhanced GRPO network architecture, which estimates the current state by computing the normalized advantage based on cumulative returns. Finally, the actor network outputs an action probability distribution, and actions are updated via importance sampling.

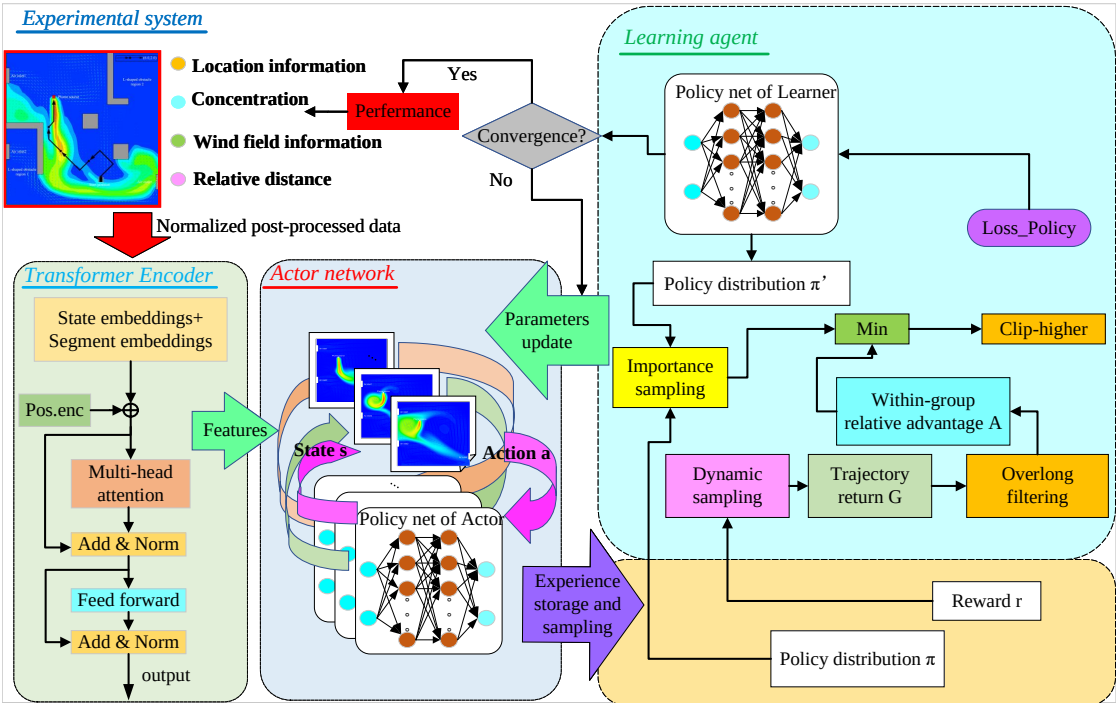


Figure 6. TOS-EGRPO algorithm diagram.

The Softmax function converts z_s into an action probability distribution. After sampling the actions with the maximum probability, the probability distribution of actions $\log P(a_t|s_t)$ is used to calculate the subsequent policy gradient. Policy updates employ the relative advantage in groups, with the update process as follows: extract the experience named $\{s_t, a_t, \log P(a_t|s_t), r_t\}$ of a round from the ReplayBuffer and compute its discounted cumulative rewards, where γ is the discount factor. The specific mathematical form is as follows:

$$G = \sum_{t=0}^{T-1} \gamma^t r_t \tag{31}$$

The mean and standard deviation are calculated by Formula (20) to obtain the standardized

group relative advantage. The objective function of actor network is calculated by Formula (23). The actor network parameters are optimized via backpropagation using gradient ascent. The update formula is as follows:

$$\theta \leftarrow \theta + \eta \nabla_\theta J_{EGRPO}(\theta) \tag{32}$$

where η represents the learning rate and $\nabla_\theta J_{EGRPO}(\theta)$ represents the gradient of objective function $J_{EGRPO}(\theta)$ with respect to the parameter θ.

After acquiring new rewards and observation information, the observation information is added to the observation queue to compute the probability distribution for the next action and make a decision. This process is repeated until the end of the round. The pseudocode for the TOS-EGRPO algorithm is summarized in Algorithm 1.

Algorithm 1 TOS-EGRPO, actor-only style

Inputs:

$$O_t = [\vec{s}_1^t, \vec{s}_2^t, \vec{s}_3^t, \cdots, \vec{s}_N^t]_{t=1,2\ldots T}, \vec{s}_i^t = [\vec{x}_i^t, \vec{y}_i^t, \vec{q}_i^t, \vec{v}_i^t, \vec{\phi}_i^t, \vec{d}_i^t]_{i=1,2\ldots N}$$

, the state observation information O_t, the policy parameter θ

Outputs: the updated parameter θ, and the state observation sequence *state_seq*.
Along with the policy network π_θ

Initialize the parameter θ of the policy network
$state_seq = queue([\mathbf{0}, \mathbf{0}, \cdots, \mathbf{0}])$
for $episode = 1, 2, \ldots, N$ **do**
  Initialize the environment and obtain the observation information O_1
  Update the queue: $state_seq = state_seq.append(O_1)$
  for $t = 1, 2, \ldots, T$ **do**
    $a_t \leftarrow \pi_\theta(state_seq)$
    Perform the actions: $O_{t+1}, r_t \leftarrow P(O_{t+1}, r_t | O_t, a_t)$
    if the buffer overflows **then**
      $\theta \leftarrow \theta + \nabla_\theta J_{EGRPO}(\theta)$
      Update the sequence:$state_seq = state_seq \cdot append(O_{t+1})$
    end if
  end for
end for

4. Results and discussion

4.1. Odor source search in diverse environments

This section investigates the impact of varying wind speeds and source release rates on the OSL task, considering the robot's starting positions at the air inlet and recirculation zones in an obstacle-free environment. The performance is evaluated in terms of accuracy and efficiency, including

average reward, average search steps, and success rate. Improved performance is characterized by faster reward convergence, fewer search steps, and a higher success rate.

The model is implemented in PyTorch, and the experimental environment consists of a Xeon Silver 4510 3.9 GHz CPU and an NVIDIA GeForce RTX 3090 GPU, running Windows 10 with PyTorch 1.12.0+cu113 and Python 3.9.

Grid resolution plays a crucial role in plume tracking and localization in the environment[46]. Due to the low computational complexity and high modeling accuracy of gas diffusion simulations, this study uses a grid resolution of 0.2 meters to get 5,000 grid cells in a 2D square space. Each grid square represents a fixed step length of 0.2 m for the robot's straight-line movement. When moving diagonally, the robot's step length is 0.28 m. The sensing interval for each step is 3 s. Due to the high speed, the robot's response time is negligible. During the training phase, all hyperparameters and empirical values used in the model are listed in Table 4. The actor network employs an Adam optimizer[47].

Table 4. Hyperparameters and training thresholds of TOS-EGRPO.

Parameter	Value
Learning rate	0.001
Hidden width	64
Entropy regularization coefficient	0.01
Batch size	512
K-epochs (Number of data reuse times)	10
Discount factor	0.99
Maximum step per episode	150
Clip lower limit	0.2
Clip upper limit	0.28
Maximum C_2H_5OH mass fraction	0.99
Maximum wind speed (m/s)	10
Maximum wind direction angle (degrees)	360
Concentration threshold for plume finding	0.05
Concentration threshold for confirming the plume source	0.6
Wind direction angle variation threshold in recirculation zones (degrees)	15
Minimum wind speed (m/s)	0.1
Safety distance threshold (m)	0.4

To investigate the sensitivity of the overall OSL performance to the threshold selection of key parameters in the PCPN, parameter sensitivity experiments for PCPN were conducted. The experimental results are presented in Table 5.

This paper defines three levels of sensitivity (high, moderate, and low) based on the degree

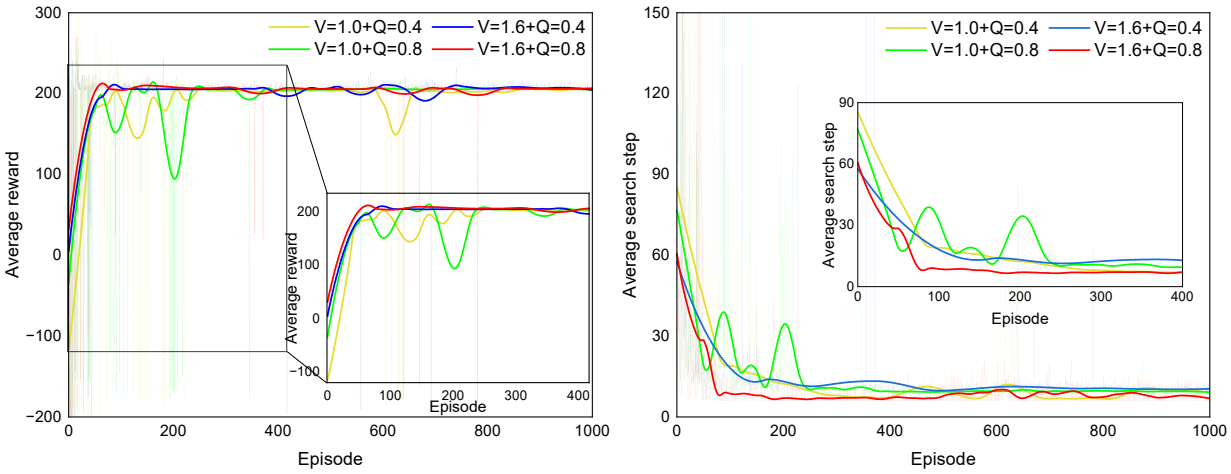
Table 5. Sensitivity of the overall OSL performance to key parameters of PCPN. Sensitivity is classified by $\Delta S = \max(S) - \min(S)$: High $\Delta S \geq 20\%$, Moderate $10\% \leq \Delta S < 20\%$, Low $\Delta S < 10\%$. Each success rate is averaged over $N = 1000$ independent evaluation.

Key parameters of PCPN	Test value	Success rate(%)	ΔS	Sensitivity
C_t	0.01	85.62	29.27%	High
	0.05	99.58		
	0.1	70.31		
$C_{\max}$	0.5	92.32	20.89%	High
	0.6	99.15		
	0.7	78.26		
$\phi_{\max}$	5	84.48	13.97%	Moderate
	15	98.45		
	30	97.98		
$U_{\min}$	0.05	96.48	1.11%	Low
	0.1	97.35		
	0.15	96.24		
d_{thr}	0.2	82.06	17.17%	Moderate
	0.4	99.23		
	0.6	98.47		

to which the success rate changes with the test value. From the Table 5, we can see that C_t and $C_{\max}$ are highly sensitive parameters whose values strongly influence the robot's ability to locate the leakage source in OSL tasks. The parameter settings adopted in this study have been empirically validated, achieving a high success rate and confirming their appropriateness. $\phi_{\max}$ and d_{thr} are moderately sensitive parameters. Both parameters exhibit good robustness when their values exceed the baseline thresholds. The values adopted in this study are selected at the lower bounds of their respective robust intervals to balance between safety and efficiency. $U_{\min}$ is a low-sensitivity parameter, indicating that the robot's ability to escape from recirculation zones depends weakly on the precise threshold value. In summary, key parameters of the TOS-EGRPO exhibit robustness within reasonable ranges.

At the initial position (1.0, 3.6) near the air inlet, the robot is guided by lateral wind to perform downwind exploration. Starting at (2.0,2.0) in the recirculation zone, the robot is influenced by recirculation during the initial phase. Based on different starting positions, we vary the wind speed and source release rate to observe the convergence process of TOS-EGRPO, as shown in Fig. 7 and Fig. 8.

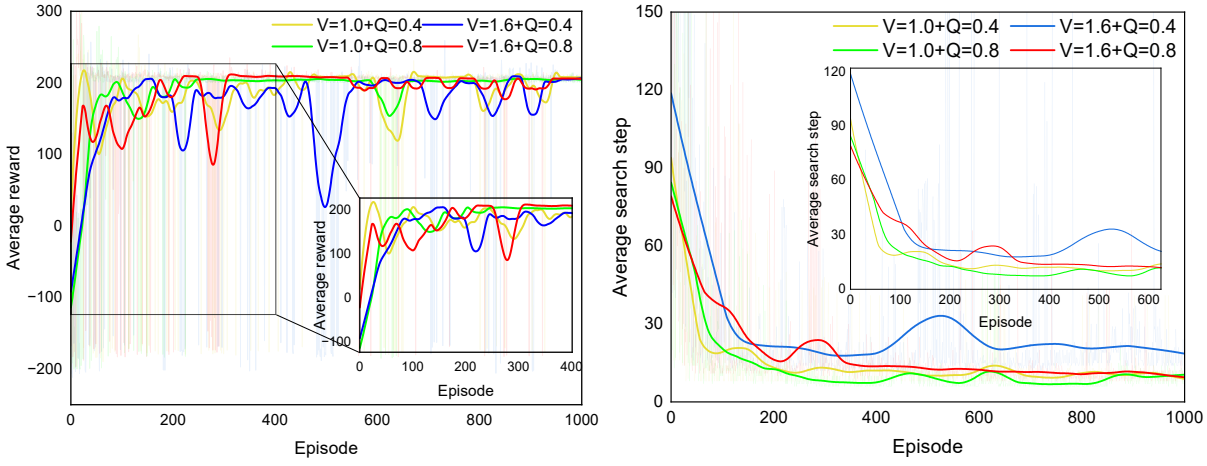
As shown in Table 6, for each configuration, the trained policy is evaluated over 50 independent episodes. The robot initialized in recirculation zones achieves a higher average reward but requires more search steps compared to the one starting at the air inlet. This is because the former must



(a) Average reward

(b) Average search steps

Figure 7. Convergence process of TOS-EGRPO under different wind speeds V and source release rate Q starting at the air inlet. The average reward is defined as the mean accumulated reward over a series of episodes, whereas the average search step refers to the average number of steps taken by the robot during multiple experimental episodes.



(a) Average reward

(b) Average search steps

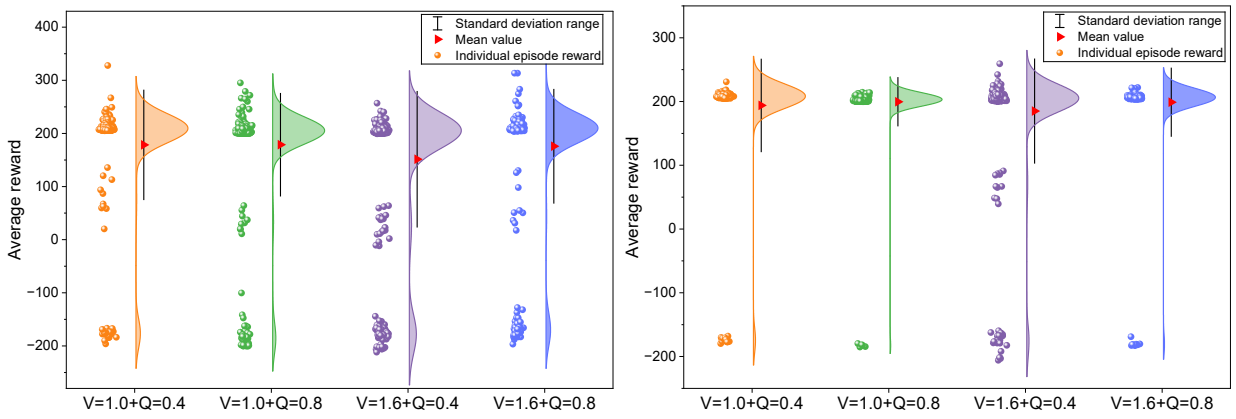
Figure 8. Convergence process of TOS-EGRPO under different wind speeds V and source release rate Q starting at the recirculation zone.

perform additional exploration to escape from local recirculation zones, whereas the latter can find the plume through straightforward downwind exploration. A comparison between the second and third experiments reveals that the robot starting in the recirculation zone obtains an average reward 21 points higher and requires 14 fewer search steps. The primary reason is that, although global wind field information guides the robot to avoid local concentration extrema, the concentration distribution still dominates the overall search process. For the success rate 98% 100%, the 95% Wilson confidence intervals are within approximately ±5% given $n = 50$.

Table 6. Search performance of TOS-EGRPO under different wind speeds V and source release rates Q at various starting positions. A=Air inlet, R=Recirculation zone.

Wind speed + Source release rate	Average reward(standard deviation)		Average steps(standard deviation)		Success rate	
	A	R	A	R	A	R
1.0m/s+0.4m³/s	189.163±74.804	186.137±90.198	13.302±19.341	14.63±18.841	99.12%	98.92%
1.6m/s+0.4m³/s	204.959±49.346	168.228±109.253	13.538±15.926	27.394±29.703	99.48%	98.38%
1.0m/s+0.8m³/s	204.164±64.158	188.615±72.402	14.351±18.199	14.571±18.057	99.53%	98.59%
1.6m/s+0.8m³/s	207.235±39.804	187.424±86.178	9.692±12.423	18.364±22.522	99.86%	98.61%

To account for the stochasticity of deep reinforcement learning, we conduct a descriptive statistical analysis and experiments with five independent random seeds (0, 8, 10, 14, and 24). The reported average reward is summarized as mean±standard deviation across the five seeds. Specifically, the experimental data from the first and the last 500 episodes are selected for statistical analysis, as shown in Figure 9. In addition, multiple random-seed experiments are performed at the air inlet, as shown in Figure 10.



(a) Average reward distribution in the first 500 episodes.

(b) Average reward distribution in next 500 episodes.

Figure 9. Descriptive statistics of TOS-EGRPO under different wind speeds and source release rates across multiple episodes. The mean, standard deviation, and individual episode rewards are shown for the first 500 and last 500 episodes.

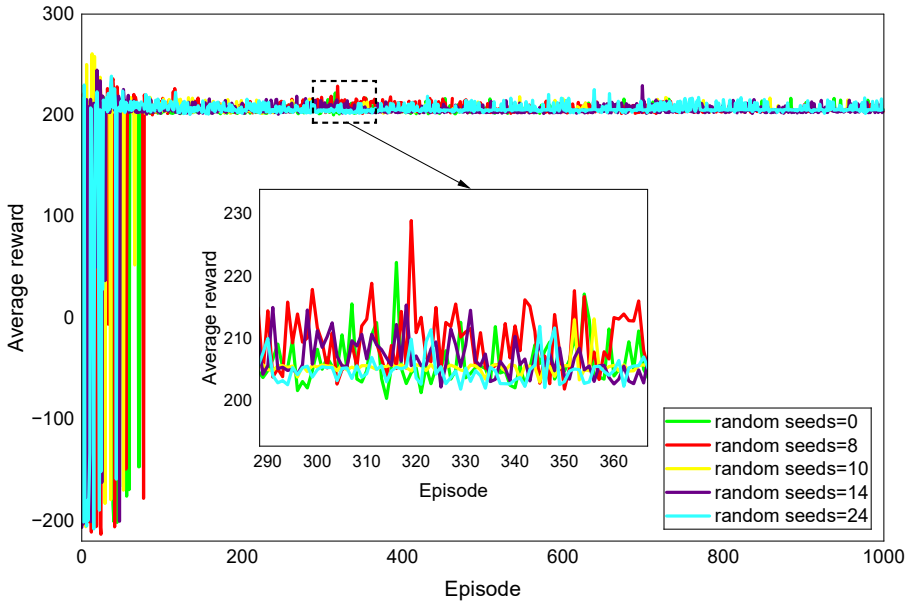


Figure 10. Average reward of the TOS-EGRPO algorithm with five random seeds

Figure 9 shows that the average rewards are relatively dispersed during the early stage of training, whereas they become more concentrated and stable in the later stage. The average reward improves progressively as the number of training episodes increases. Figure 10 demonstrates the training curves for five random seeds. The aggregated average reward is 193.90±13.50, confirming the stability of TOS-EGRPO across different random initializations.

4.2. Wind guidance analysis in the environment with obstacles

In the environment with obstacles, the number of training episodes is increased by 200 compared to obstacle-free settings. In addition, according to Formula (12), the robot incurs greater penalties as it approaches obstacles. Under identical hyperparameter settings, the robot is initialized at two positions in the wind field, (6.0, 2.0) and (6.0, 4.0), and at two positions in the calm zone, (7.0, 5.0) and (9.0, 6.0), as shown in Fig. 11. The average reward and average search steps for robots with and without wind guidance in environments with obstacles are shown in Fig. 12.

The robot starting at (6.0, 4.0) exhibits more frequent reward reductions than the one at (6.0, 2.0). The main reason is that it is closer to obstacles which leads to more frequent penalties during early exploration. However, the robot starting at (9.0, 6.0) displays reward fluctuations until after episode 1100. This phenomenon indicates that the robot without wind guidance enters the L-shaped obstacle region during initial plume exploration. Consequently, the failure rate increases due to repeated collisions and boundary violations, as the robot can only rely on accumulated experience to escape. The performance of the robot at different initial positions in OSL tasks is

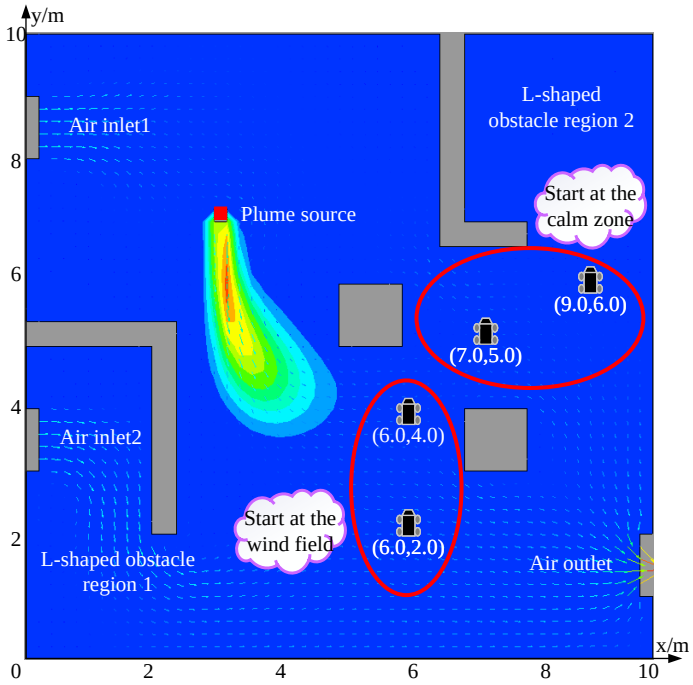
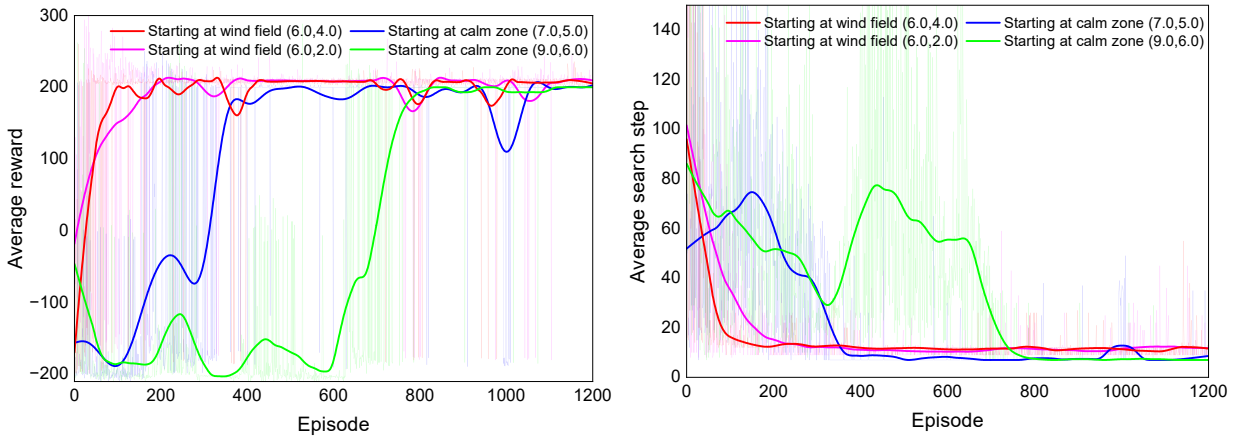


Figure 11. Starting from different positions in the environment with obstacles.



(a) Average reward

(b) Average search steps

Figure 12. TOS-EGRPO convergence process with or without wind guidance at the initial position in an environment with obstacles.

summarized in Table 7.

Table 7. Search performance of the robot at the initial position with or without wind guidance in the environment with obstacles.

	Starting position	Average reward	Average steps	Standard deviation of average reward	Standard deviation of average steps	Success rate
With wind	(6.0,2.0)	192.719	16.576	80.328	20.157	99.75%
	(6.0,4.0)	186.773	14.838	86.767	15.906	99.59%
No wind	(7.0,5.0)	107.248	21.188	163.906	29.461	82.53%
	(9.0,6.0)	-14.204	36.548	193.742	39.874	64.27%

As shown in Table 7, the robot initialized at wind-guided position (6.0,2.0) achieves an average reward 85 points higher and requires 5 fewer search steps compared to the one at (7.0,5.0) without wind guidance, while both maintain a success rate of approximately 99%. The relatively small standard deviations of average reward and average search steps in the two wind-guided scenarios indicate reduced variability in search behavior and more stable reward acquisition. This demonstrates that directional cues provided by the global wind field effectively decrease the randomness in plume search. As shown in Fig. 13, the robots starting from two different initial positions are able to successfully locate the plume source.

4.3. Analysis of attention weights

To study the feature aggregation performance of State token, an ablation comparative study is conducted between State token and other aggregation strategies, including last-token, mean pooling, State token in the end, and State token at the beginning. The experimental results are shown in Fig. 14 and Table 8 . Reward and search steps are represented by mean standard deviation. Convergence is declared at the first training episode where the success rate over 20 consecutive episodes reaches at least 95% and the moving-average reward falls within ±2% of its final converged value.

Table 8. Search performance of the robot in different feature aggregation strategies

Feature aggregation strategy	Reward	Search steps	Convergence episode
last-token	187.276 ± 78.582	14.129 ± 25.05	497
State token in the end	186.71 ± 84.079	11.051 ± 15.896	793
mean pooling	191.125 ± 74.281	14.66 ± 23.089	305
State token at the beginning	202.394 ± 37.029	8.912 ± 14.421	129

As shown in Fig. 14 and Table 8, all four feature aggregation methods are able to converge, but last-token, mean pooling, and State token in the end exhibit varying degrees of decline in average reward. In contrast, State token at the beginning not only converges rapidly but also remains stable

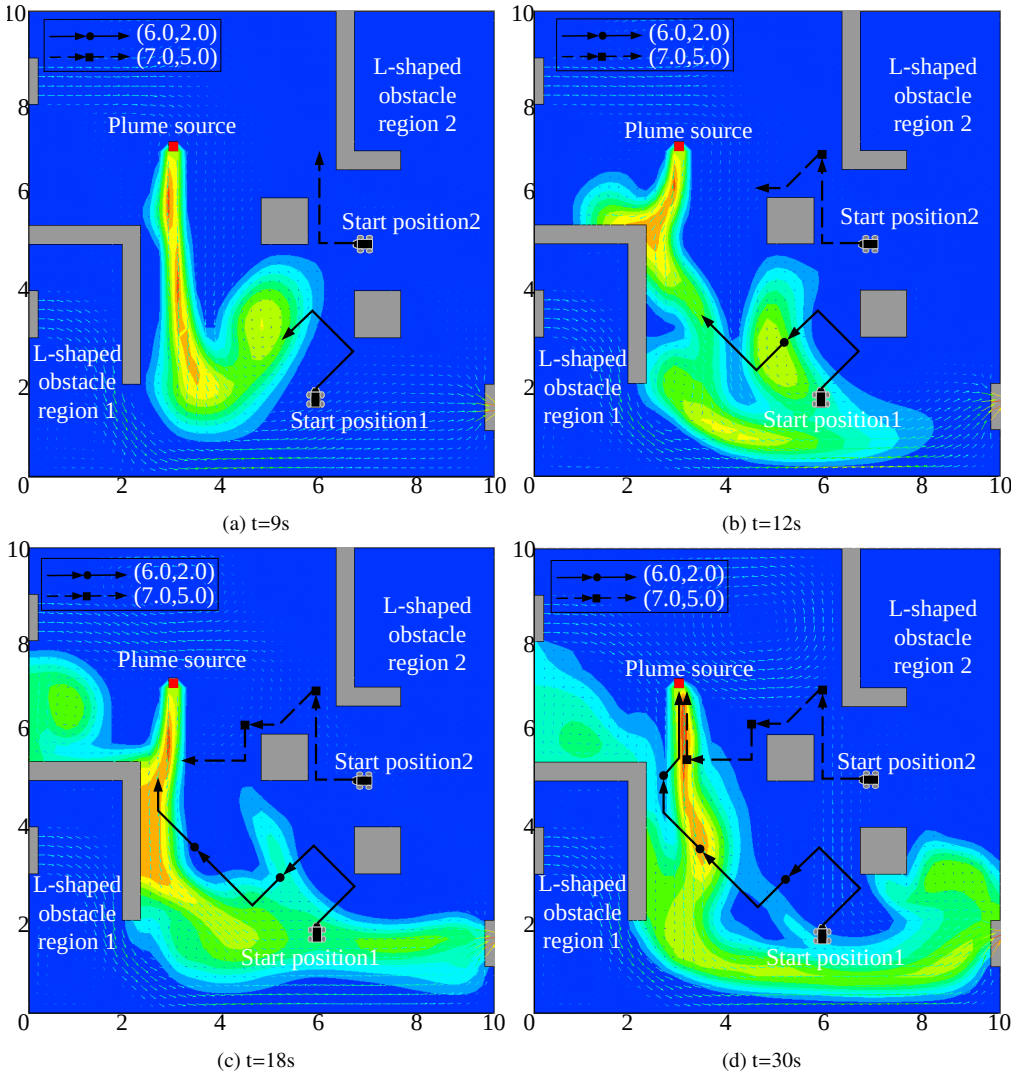


Figure 13. Search process of the odor source at different times at the initial positions (6.0,2.0) and (7.0,5.0) in the environment with obstacles.

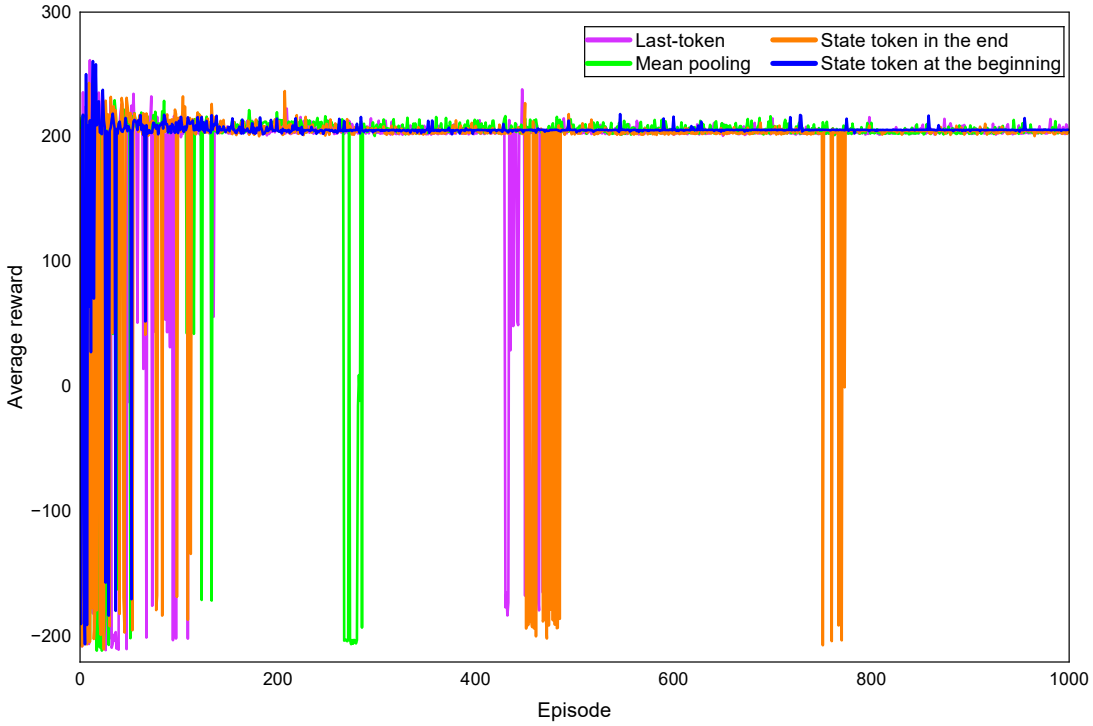


Figure 14. Average reward under different feature aggregation strategies.

in the later stages of training, indicating its strong capability in feature aggregation for enhancing overall OSL performance.

To investigate the attention allocation mechanism of the Transformer encoder under multi-feature input conditions, this study analyzes five representative scenarios in the OSL task: downwind exploration, recirculation zone initiation, upwind exploitation, obstacle avoidance, and source confirmation. Four key input features are considered, including concentration q, wind speed v, wind direction ϕ, and relative distance d. By combining these features with the reward weight assigned to each scenario, the study investigates how the Transformer's attention mechanism guides the robot's decision-making process, as shown in Fig. 15.

The attention weight calculation based on the approach by Fan et al. [48] is modified: firstly, the similarity between queries and keys of each feature in the encoder layer is calculated; subsequently, the average weights are normalized to sum to 1; finally, the detailed proportion of attention weights for each feature is derived. For each scenario, the two features with the highest weight proportions are selected and their rewards are evaluated and normalized into eight intervals, as illustrated in Fig. 15.

In the downwind exploration scenario, the reward weight for wind field information progressively increases, while the reward for relative distance exhibits pronounced fluctuations. This

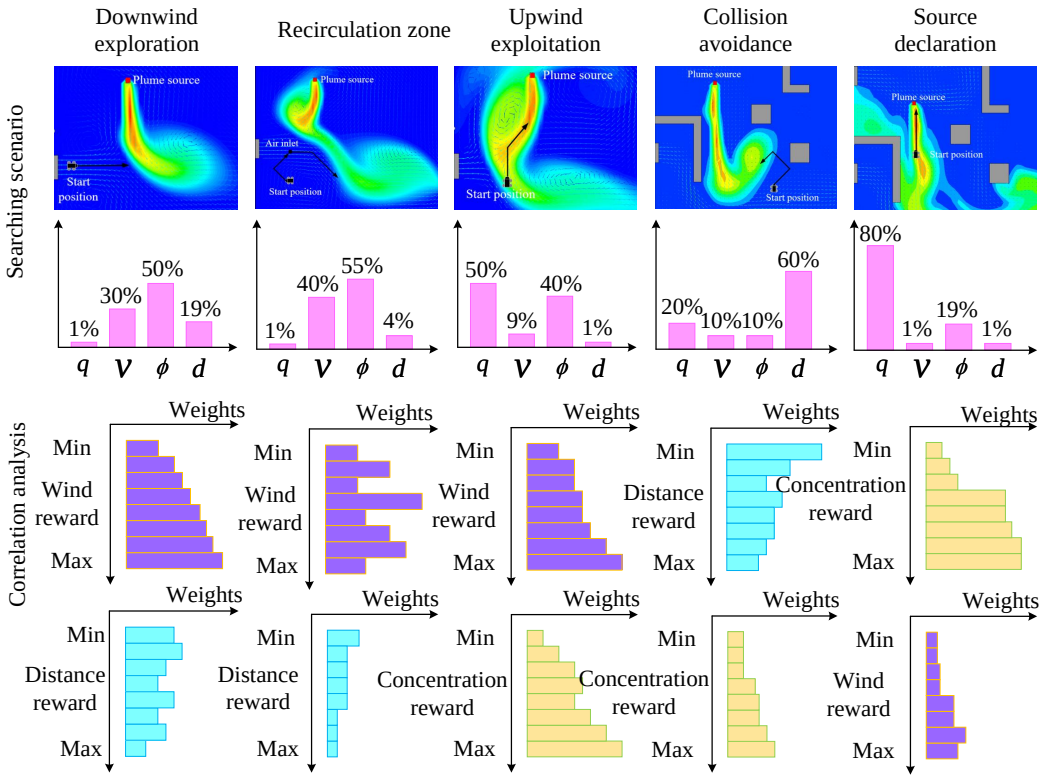


Figure 15. Visualization of Transformer encoder weights for different feature inputs in different search phases.

indicates that during the initial phase, the robot relies predominantly on wind cues to find the plume and often exceeds boundaries, but it gradually achieves downwind exploration and reduces boundary violations through training. In obstacle avoidance and source confirmation scenarios, the robot is assigned higher attention to the minimum relative distance and maximum concentration values, respectively. These results demonstrate that TOS-EGRPO effectively mitigates feature sparsity in RL algorithms by integrating attention distributions with reward weights.

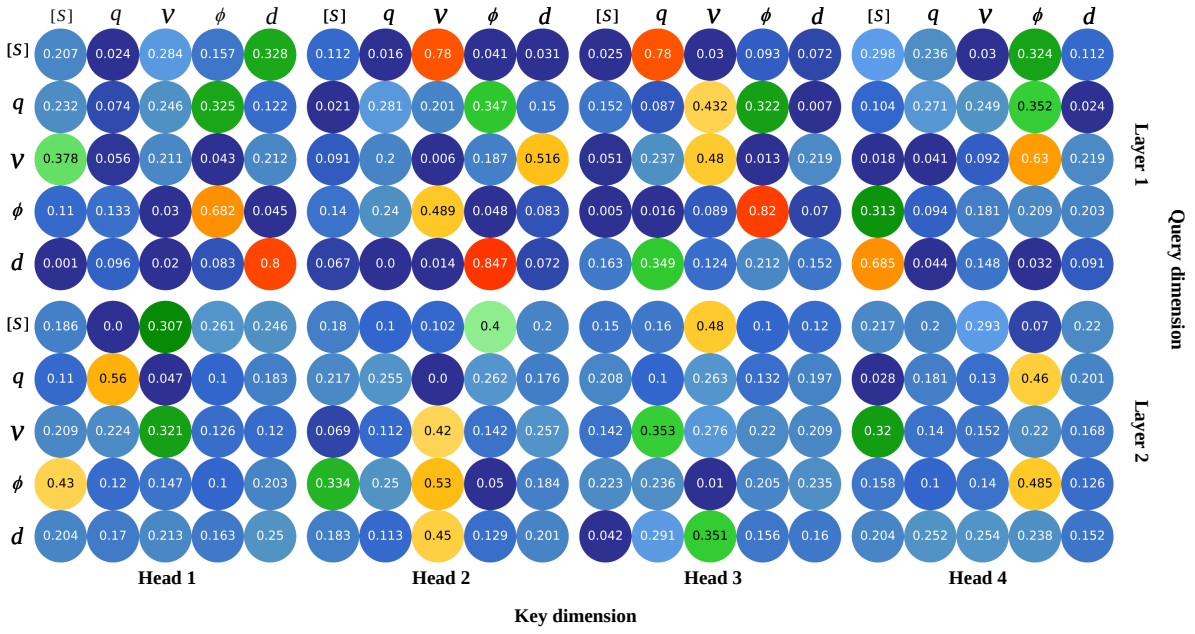
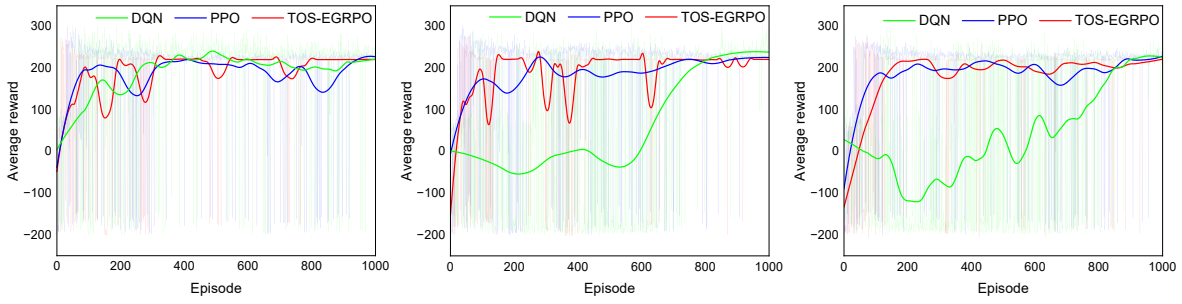


Figure 16. Visualization of attention maps in different layers of Transformer encoder, where [S] represents State token.

As illustrated in Fig. 16, this study utilizes a two-layer Transformer encoder equipped with a four-head self-attention module. Visualization analyses conducted in the environment with obstacles reveal average attention weight variances of 0.0414 and 0.0125 for the first and second layers, respectively, and average cross-head similarities of 0.4878 and 0.7547. The variance is the average variance over all attention elements, and the similarity is the average dot product between attention heads within each layer. The declining variance across layers indicates the progressive extraction of critical information and enhanced stability in decision-making. Meanwhile, the increasing cross-head similarity suggests that feature representations become more convergent in deeper layers. The main reason is the dominance of features like concentration and wind field information in the feature representation. This phenomenon is consistent with observed perceptual characteristics in practical OSL scenarios.

4.4. Comparison with other methods

In this section, we evaluate the robot's capability to handle state information loss and process long sequences, comparing the TOS-EGRPO algorithm with PPO and DQN. Experiments are conducted in an obstacle-free environment, where Poisson noise is added to normalized data, and the robot's initial position is set at (1.0, 3.6). We examine performance under varying observation sequence lengths and state information loss rates. The training results are presented in Fig. 17 and Fig. 18.



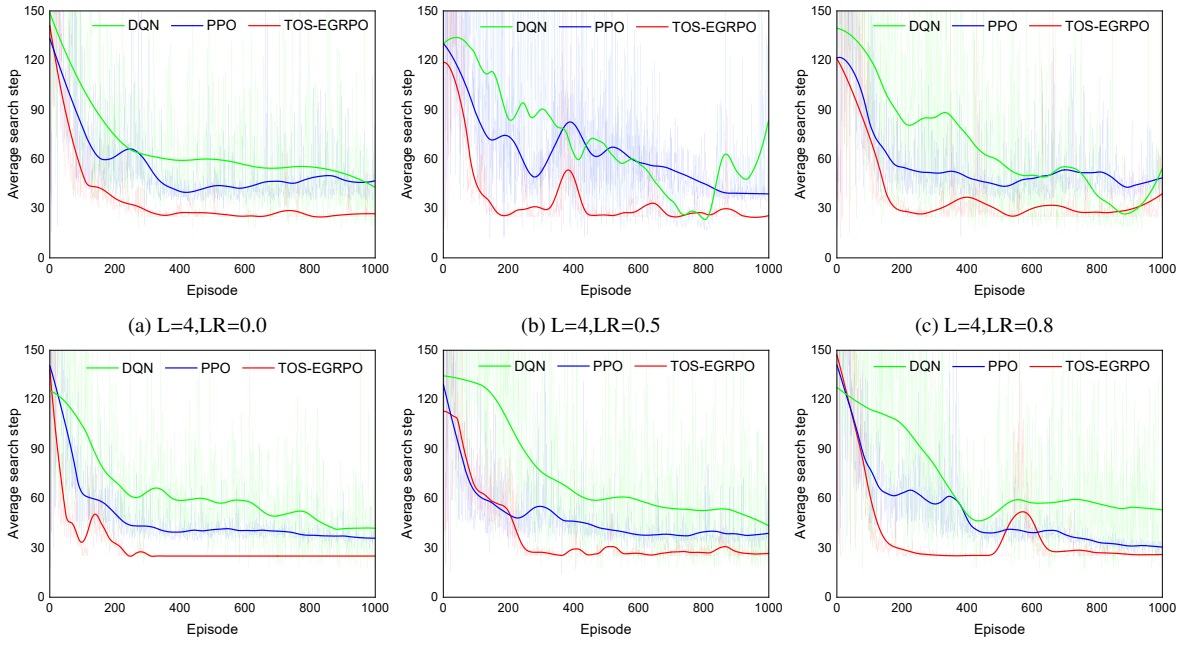
(a) L=4,LR=0.0 (b) L=4,LR=0.5 (c) L=4,LR=0.8

(d) L=8,LR=0.0 (e) L=8,LR=0.5 (f) L=8,LR=0.8

Figure 17. Average reward in the wind-guided and obstacle-free environment, where L represents the length of the observation sequence and LR represents the state information loss rate.

As shown in Fig. 17 and Fig. 18, the performance of all three algorithms deteriorates as the state information loss rate increases. However, TOS-EGRPO remains robust when the loss rate is below 0.5. Once the loss rate exceeds this threshold, DQN exhibits severe performance fluctuations and fails to converge, whereas PPO converges more slowly than TOS-EGRPO. Without the state information loss, TOS-EGRPO with longer observation sequences demonstrates better search performance. The experimental results demonstrate that TOS-EGRPO exhibits stronger robustness to state information loss compared to DQN and PPO, while effectively leveraging sequential features to mitigate noise interference.

As shown in Table 9, the success rate of DQN decreases as the state information loss rate increases for a fixed observation sequence length, making it difficult for the robot to complete the



(a) L=4,LR=0.0 (b) L=4,LR=0.5 (c) L=4,LR=0.8

(d) L=8,LR=0.0 (e) L=8,LR=0.5 (f) L=8,LR=0.8

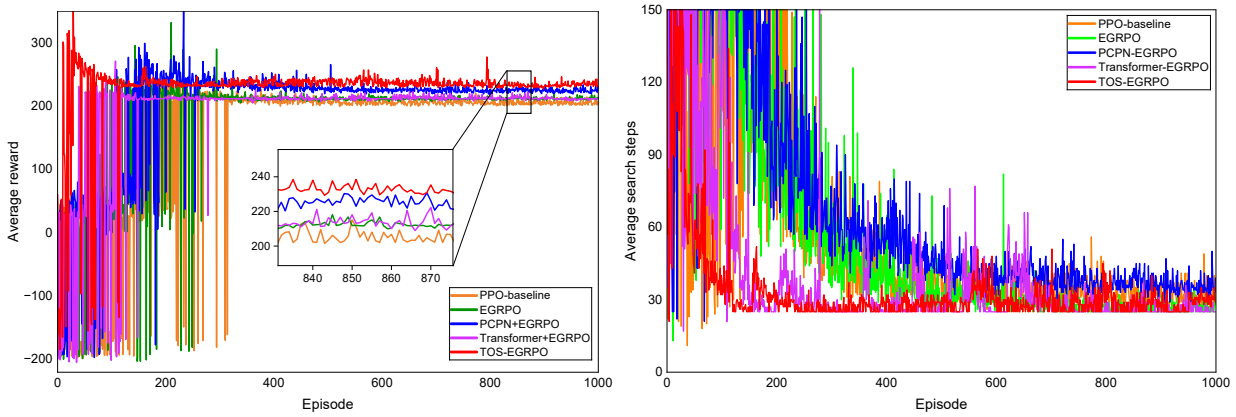
Figure 18. Average search steps in the wind-guided and obstacle-free environment.

Table 9. Performance comparison of RL algorithms for OSL tasks in the windy and obstacle-free environment.

	Average reward			Average search steps			Success rate		
L=4	LR=0.0	LR=0.5	LR=0.8	LR=0.0	LR=0.5	LR=0.8	LR=0.0	LR=0.5	LR=0.8
DQN	189.151	61.728	40.247	67.051	71.463	68.869	97.21%	12.42%	10.04%
PPO	187.293	186.542	186.474	55.256	63.606	57.044	97.53%	95.83%	93.21%
Ours	193.497	190.704	184.660	35.147	36.454	39.03	98.45%	98.18%	97.25%
L=8	LR=0.0	LR=0.5	LR=0.8	LR=0.0	LR=0.5	LR=0.8	LR=0.0	LR=0.5	LR=0.8
DQN	179.593	160.896	113.732	63.834	74.58	71.935	97.62%	82.48%	40.37%
PPO	202.881	196.285	196.623	48.264	48.442	51.329	98.36%	97.14%	97.08%
Ours	206.464	197.343	196.431	31.089	38.124	37.997	99.75%	98.75%	98.24%

OSL task when the loss rate exceeds 0.5. In contrast, PPO is more sensitive to observation sequence length. For example, when the loss rate rises from 0.5 to 0.8, the success rate decreases by 2% for a sequence length of 4, whereas it declines by only 0.6% for a sequence length of 8. Although TOS-EGRPO is influenced by both sequence length and information loss rate, it consistently achieves the highest success rate. When the sequence length is 8, success rate fluctuations for TOS-EGRPO and PPO remain in 1%, yet TOS-EGRPO maintains higher average rewards and fewer search steps. This demonstrates that TOS-EGRPO effectively utilizes information from long observation sequences and uses the Transformer encoder to process time-series information to make better decisions. Otherwise, DQN shows a pronounced decline in success rate with increasing information loss, regardless of sequence length, indicating its limited capacity to exploit long-sequence observations.

To investigate the overall performance of the TOS-EGRPO algorithm, this paper employs PPO as the baseline and conducts an ablation study by removing each component to clarify their respective contributions. Experiments are carried out under both obstacle-free conditions and scenarios involving Poisson noise and information loss, with the results presented in Figure 19.



(a) Average reward

(b) Average search steps

Figure 19. TOS-EGRPO's OSL performance in different ablation conditions.

As shown in Figure 19, compared with the PPO baseline, PCPN facilitates an increase in the reward obtained by the robot. This indicates that, under the guidance of PCPN, the robot makes decisions on critical events and concentrates more on downwind exploration and upwind exploitation in the OSL task, thereby achieving higher rewards. Transformer-based feature extraction enables the robot to rapidly learn key features in the presence of Poisson noise and information loss, resulting in fast convergence. By integrating the advantages of both approaches, TOS-EGRPO completes the OSL task with the fastest convergence, the highest average reward, and the minimum number of search steps.

To compare the search performance in OSL tasks with non-RL algorithms, experiments are conducted in both obstacle-free and obstacle-present environments, with the initial position at (8.0,1.6). Poisson noise, state information loss, and sequential observation are considered in the experiment (L=8, LR=0.5). The TOS-EGRPO algorithm is compared with the hex-path algorithm[49](gradient-based), Bayesian inference[50], shark smell optimization[51], and Clutaxis[52] (information entropy-based). Experimental results are presented in Table 10.

Table 10. Performance comparison of Non-RL algorithms for OSL tasks in different environments. N=No obstacles, W=With obstacles.

Methods	Average search steps		Average search distance(m)		Success rate	
	N	W	N	W	N	W
Gradient-based	75.048	96.251	19.714	25.084	64.27%	45.02%
Bayesian inference	67.835	80.106	17.315	20.789	75.41%	48.38%
SSO	56.386	72.141	14.94	18.622	82.66%	70.85%
Clutaxis	49.257	62.482	12.348	16.058	92.46%	86.25%
TOS-EGRPO	23.415	27.854	6.369	7.604	98.75%	97.86%

As shown in Table 9, the gradient-based algorithm achieves lower success rates than other methods in both experimental environments, indicating limited search performance. In addition, both Bayesian inference and SSO algorithms are less robust to environmental variations compared to TOS-EGRPO. The Clutaxis algorithm based on information entropy achieves a success rate comparable to TOS-EGRPO, with both exceeding 90%. However, Clutaxis requires more average search steps in obstacle-present environments. The main reason is that entropy-maximizing algorithms encourage the robot to perform broader exploration. This tendency increases the likelihood of non-productive regions, thereby increasing the average search steps and distance. In other words, TOS-EGRPO's OSL performance in dealing with information loss and Poisson noise outperforms other non-RL methods.

In summary, the TOS-EGRPO algorithm demonstrates efficient search performance in obstacle-free environments with both state information loss and noise interference, as well as in complex environments with obstacles.

5. Conclusion

This paper proposes a soft computing approach in OSL tasks, termed TOS-EGRPO, which integrates a PCPN-based Transformer encoder with an enhanced GRPO. The proposed approach addresses two problems: insufficient exploration in the OSL process leads to the robot being trapped in recirculation zones and local optima, and noise interference in practical environments causes partial sensor information loss. TOS-EGRPO demonstrates high search performance in key events

such as downwind exploration, upwind exploitation, collision avoidance, and source declaration. The plume source localization mechanism based on PCPN is designed to enable dynamic switching among different search strategies, enhancing the robot's exploration capabilities. Considering realistic environmental challenges such as noise interference, the OSL problem is formulated as a POMDP. TOS-EGRPO employs the Transformer encoder with the State token to extract and combine multiple feature types (e.g., concentration distributions, wind-field information, and relative distances), adaptively allocating attention to heterogeneous inputs while processing observation sequences in parallel. OSL experiments show that the TOS-EGRPO outperforms the PPO and DQN in terms of average search steps, training efficiency, and success rate. Under various event scenarios, TOS-EGRPO outperforms the baseline PPO, increasing the average reward by 14.4% and reducing average search steps by 28.5%. Compared to the non-RL methods, TOS-EGRPO demonstrates fewer average search steps and higher success rate. These experimental results confirm its robustness against interference and enhanced sequence-processing capabilities.

However, the CFD simulation experiments in this study are currently restricted to a 2D space, with the search scenarios limited to a single odor leakage source and a relatively small environmental scale. Furthermore, a time-varying periodic source is adopted for the release rate, which exhibits limited generalization capability to real-world gas leakage patterns. Therefore, our future work will focus on extending the algorithm to 3D environments and incorporating more complex gas leakage patterns to improve the generalizability of the developed policies. Moreover, a heterogeneous cooperative framework combining Transformer-based feature extraction and PCPN will be developed to improve environmental perception and enable efficient multiple odor source localization.

CRediT authorship contribution statement

Qing Li: Methodology, Formal analysis, Investigation, Writing - original draft, Project administration, Writing - review & editing. **Jie Yuan:** Data curation, Writing - review & editing, Funding acquisition, Software. **Li Zhu:** Investigation, Software. **Ningning Zhang:** Investigation, Software. **Erkenbieke Jia:** Data curation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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